# Martin Cooler-Stith

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## EDUCATION

**Bachelor of Electrical Engineering** - Expected May 2028
College of Science and Engineering, University of Minnesota Twin Cities
GPA: 3.5/4.0

Relevant Coursework Completed: Signals, Circuits & Electronics; Analog & Digital Electronics; Introduction to Digital System Design; Introduction to Microcontrollers

Current Coursework: Signals & Systems; Circuits & Electronics Lab I; Analog Electronics; Semiconductor Devices

## TECHNICAL SKILLS

**Lab & prototyping:** Oscilloscope, soldering, circuit debugging, 3D printing, laser-cut fabrication
**Robotics & embedded systems:** PIC24, ESP32, ultrasonic sensors, PWM motor control
**Data & programming:** Python, MATLAB, C++
**Design tools:** Fusion 360, Onshape

## PROJECT EXPERIENCE

**Research Assistant; Reinforcement Learning for Quadcopters** | Prof. Nicola Elia | May 2026-Present
- Built containerized workflows for quadcopter reinforcement-learning training and integrated Google Cloud services to support training runs.
- Designed and built a low-cost tachometer to measure drone-propeller angular velocity in a research lab.
- Designed and fabricated a mount for in-place quadcopter testing using 3D-printed and laser-cut components.
- DOE with tachometer, recorded test measurements, analyzed results in Python, and documented findings for Prof. Elia.

**Maze-Solving Robot** | EE 2361: Introduction to Microcontrollers | 2026
- Developed PIC24 firmware for a maze-solving robot using three ultrasonic sensors for navigation.
- Implemented PWM control of two brushed DC motors for speed-controlled movement.
- Diagnosed hardware issues using an oscilloscope and tested sensor and motor behavior in the lab.

**Mesh Network** | 2024
- Developed and implemented a network of three ESP32 nodes using ESP-NOW protocol.
- Performed range testing on nodes to determine signal area.

## LEADERSHIP EXPERIENCE

**Community Manager** | Beacon Social Club | Nov. 2025-Present
- Resolved a conflict between community members by talking with them about the situation and helping them reach a peaceful resolution.

**CAD Lead** | FIRST Tech Challenge Team #14575, Capital Bots | 2021-2023
- Led the team's Fusion 360 mechanical design.
- Mentored new members as they learned robotics.

## VOLUNTEER EXPERIENCE

**Volunteer Intern** | KID Museum, Bethesda, MD | Sep. 2022-May 2023
- Assisted children with Snap Circuits projects, taught technical skills, and encouraged their interest in science & engineering.
- Helped children learn to code with Scratch and supported their creativity.

**Volunteer Intern** | Greenbelt Access Television, Greenbelt, MD | Sep. 2021-May 2022
- Set up lights and props for local film projects and operated a camera.
- Collaborated with the director.

**Volunteer Intern** | The SPACE Free Art for All, Greenbelt, MD | Sep. 2020-May 2021
- Tutored students in Scratch and produced recorded Python programming lessons with custom animations.
- Welcomed visitors to the makerspace.